Delta Physics

Select a topic of your Choice

5 minutes – Individual

Special Relativity

A muon lives for 2.2 μs. How do so many reach the ground?



Higher Physics

PART 2

Time Dilation

A clock flies round the world. Does it still agree with a clock on the ground?

By the end of this part…



1. I can **describe** how a **light clock** shows **time dilation**
2. I can **use** $t' = t / \sqrt{(1 - (v/c)^2)}$ to find **t'** or **t**
3. I can **state** that **moving clocks run slow** for a stationary observer
4. I can **explain** why **everyday speeds** show **no** time dilation

The Light Clock

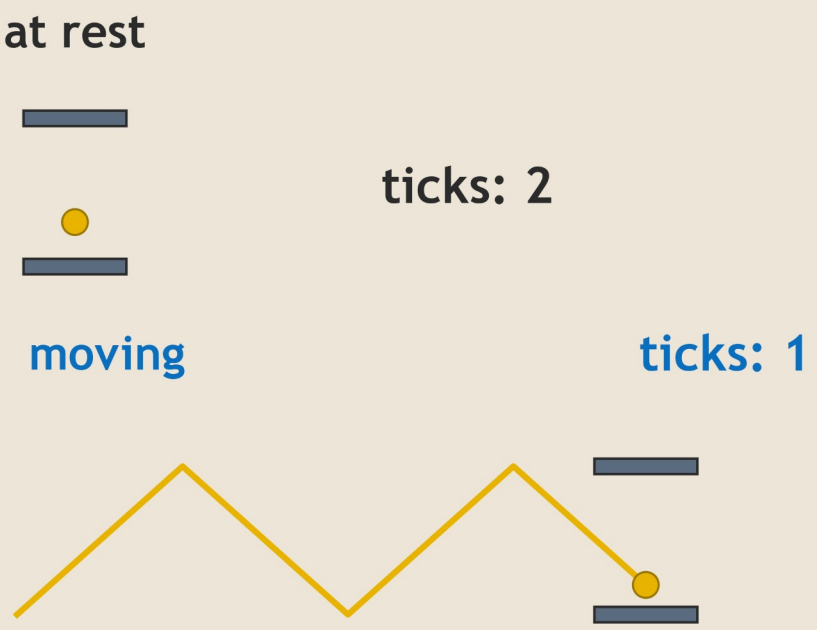
Tick

light goes **up and back** between two mirrors

Moving

seen from outside, the light goes **further**

Try it: the light clock on the website.



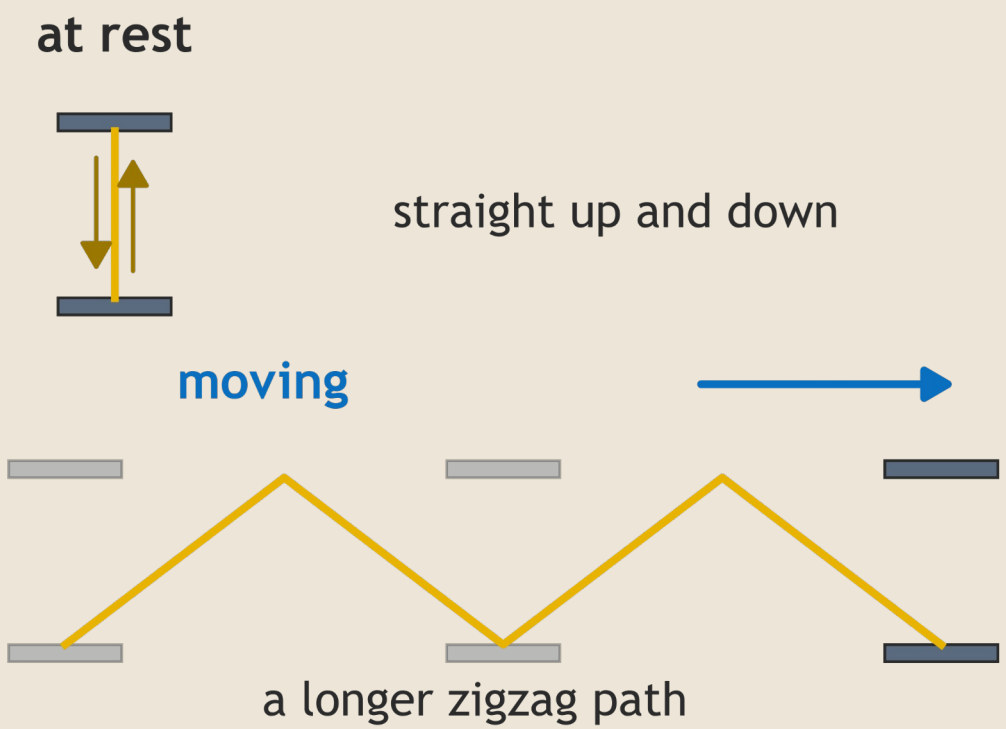
Moving Clocks Run Slow

c is fixed

a **longer path** takes **longer**

So

a **moving clock** runs **slow** for a **stationary observer: time dilation**



t and t′

t

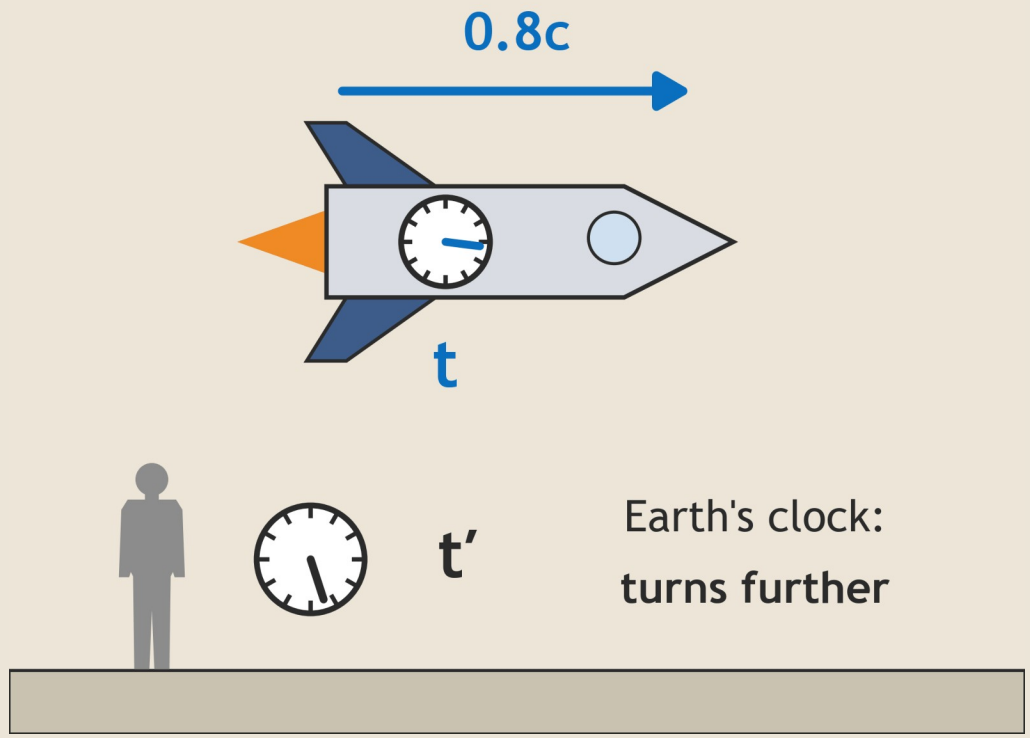
time in its **own frame**: the clock **moves with** it

t′

time for the **other observer**

Check

t′ is longer than t



Time Dilation

$$t' = t/\sqrt{(1 - (v/c)^2)}$$

t′

other observer's time (s)

t

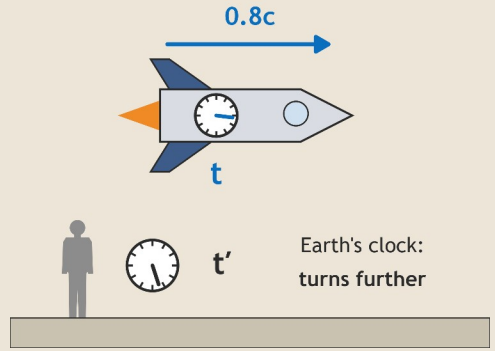
own frame time (s)

v

speed (m s^{-1})

c

3.00×10^8 m s^{-1}



Worked Example 1

A muon has a mean lifetime of 2.2×10^{-6} s in its own frame of reference. It travels at 0.98c relative to an observer on Earth.

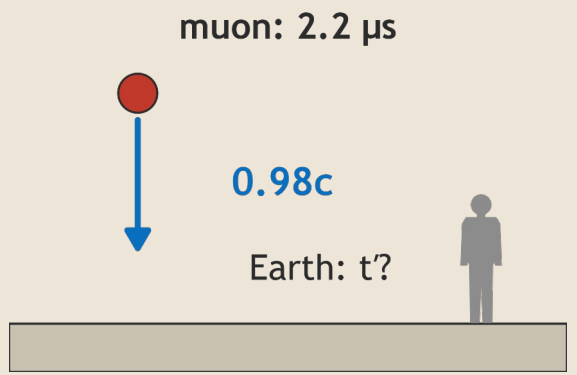
Calculate the mean lifetime of the muon measured by the observer on Earth. **3**

$$t' = t \,/\, \sqrt{(1 - (v/c)^2)}$$

$$t' = 2.2 \times 10^{-6} \,/\, \sqrt{(1 - (0.98c/c)^2)}$$

$$t' = 1.1 \times 10^{-5} \text{ s}$$

3 marks: relationship (1), substitution (1), answer with unit (1).



Worked Example 2

A spacecraft travels between two planets at 0.6c relative to them. An observer on one of the planets measures the journey time as 50 s.

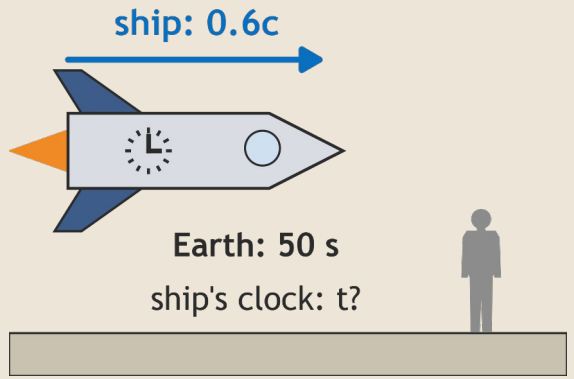
Calculate the journey time measured by a clock on the spacecraft. **3**

$$t' = t \,/\, \sqrt{(1 - (v/c)^2)} \quad \text{(planet's 50 s is } t')$$

$$50 = t \,/\, \sqrt{(1 - (0.6c/c)^2)}$$

$$t = 40 \text{ s}$$

3 marks: relationship (1), substitution (1), answer with unit (1).



Your Turn 1

A spacecraft travels at 0.5c relative to Earth. A clock on the spacecraft measures a time interval of 30 s.

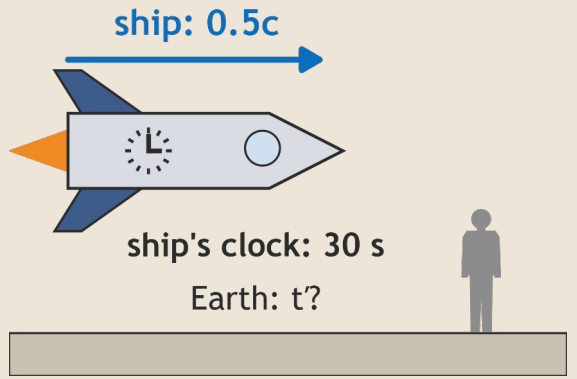
Calculate the time interval measured by an observer on Earth. **3**

$$t' = t \,/\, \sqrt{(1 - (v/c)^2)}$$

$$t' = 30 \,/\, \sqrt{(1 - (0.5c/c)^2)}$$

$$t' = 34.6 \text{ s}$$

3 marks: relationship (1), substitution (1), answer with unit (1).



Your Turn 2

A spacecraft travels between two planets at 0.8c relative to them. An observer on one of the planets measures the journey time as 100 s.

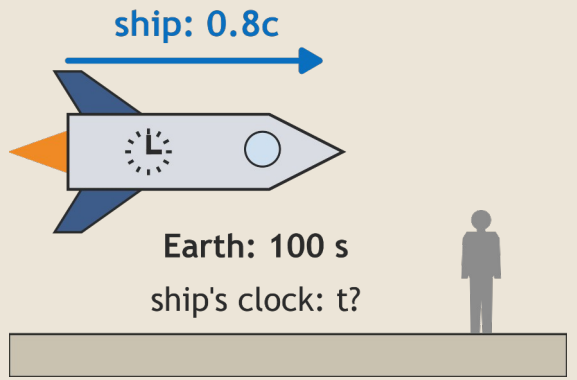
Calculate the journey time measured by a clock on the spacecraft. **3**

$t' = t / \sqrt{(1 - (v/c)^2)}$ (planet's 100 s is t′)

$100 = t / \sqrt{(1 - (0.8c/c)^2)}$

$t = 60\ s$

3 marks: relationship (1), substitution (1), answer with unit (1).



Past Paper: 2025 Paper 2 Q4(a)(b)

4. The Theory of Special Relativity states that the measurement of time for an observer moving at high speed is changed relative to a stationary observer.

(a) State the name of this effect. **1**

(b) A spacecraft is travelling at a constant speed of 2.29×10^8 m s^{-1} relative to a nearby planet.

A technician on the spacecraft measures a time interval to be 245 s.

Calculate the time interval as measured by an observer on the nearby planet. **3**

Space for working and answer

Scan: page 14.

Plan: the **technician's** 245 s is **t**

Marking: 2025 Paper 2 Q4(a)(b)

Question			Expected response	Max mark	Additional guidance
4.	(a)		Time dilation	**1**	
	(b)		$t' = \dfrac{t}{\sqrt{1-\left(\dfrac{v}{c}\right)^2}}$ (1) $t' = \dfrac{245}{\sqrt{1-\left(\dfrac{2.29\times10^8}{3.00\times10^8}\right)^2}}$ (1) $t' = 379$ s (1)	**3**	Accept: 380, 379.3, 379.25 Accept: $t' = \dfrac{245}{\sqrt{1-\left(\dfrac{2.29}{3.00}\right)^2}}$

Past Paper: 2024 Q7

7. A spacecraft is travelling at a speed of $0.20c$ relative to the Earth.

The spacecraft emits a signal for 20.0 s as measured in the frame of reference of the spacecraft.

An observer on Earth measures the duration of the signal as

A 19.2 s

B 19.6 s

C 20.0 s

D 20.4 s

E 20.8 s.

Answer D: $t' = 20.0 / \sqrt{(1 - 0.20^2)} = 20.4$ s.

Scan: page 7.

Past Paper: 2023 Q7

7. Muons are created in the upper atmosphere of the Earth.

The mean lifetime of these muons in their frame of reference is 2.20 μs.

The muons are travelling at $0.99c$ relative to an observer on Earth.

The observer measures the mean lifetime of these muons as

A 1.56×10^{-2} s

B 2.20×10^{-3} s

C 1.11×10^{-4} s

D 1.56×10^{-5} s

E 3.10×10^{-7} s.

Answer D: $t' = 2.20 \times 10^{-6} / \sqrt{(1 - 0.99^2)} = 1.56 \times 10^{-5}$ s.

Scan: page 7.

Past Paper: 2021 Q10

10. A spacecraft is travelling at a speed of $0{\cdot}45c$ relative to Earth.

An observer on Earth measures the time taken for the spacecraft to travel between two points to be 72 hours.

An observer on the spacecraft measures the time taken to travel between these two points to be

A 53 hours

B 64 hours

C 72 hours

D 81 hours

E 90 hours.

Answer B: 72 h on Earth is t′. t = 72 × √(1 − 0.45²) = 64 h.

Scan: page 8.

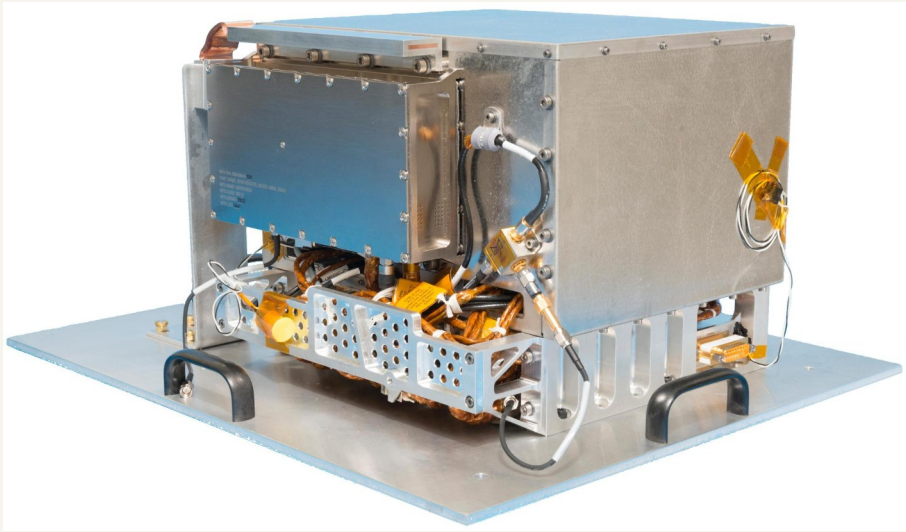
Evidence: Atomic Clocks

1971: atomic clocks flown round the world

Back home, they **disagreed** with clocks on the ground, as **relativity** predicts

Today, atomic clocks fly in **space**



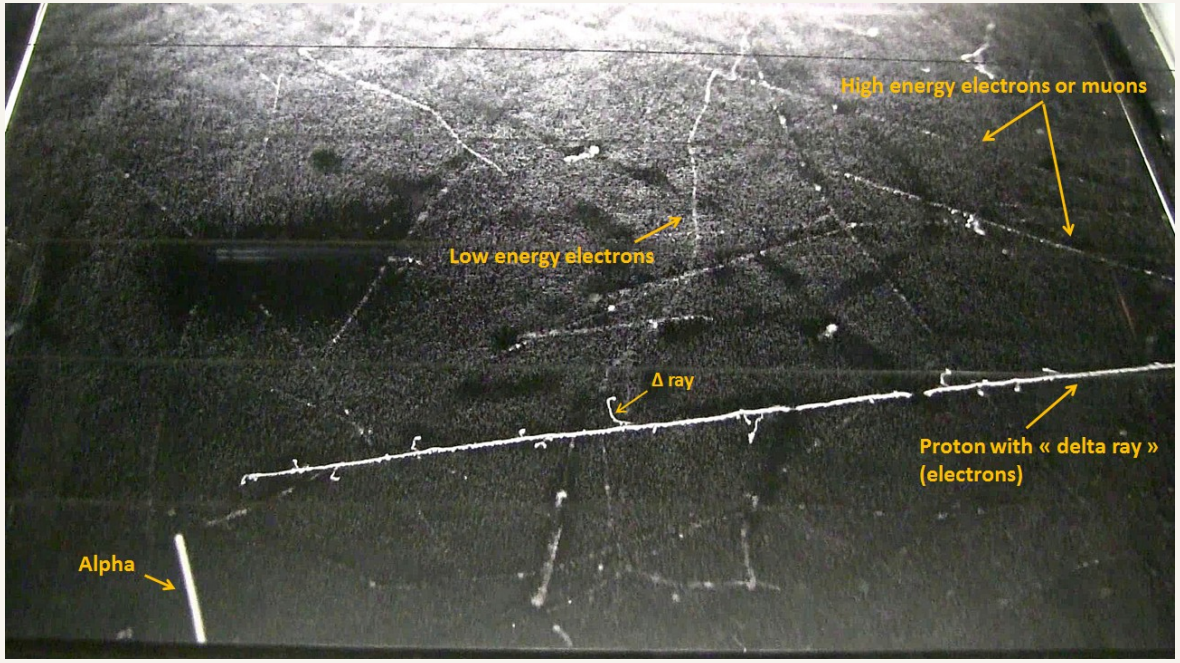


Evidence: Muons

Muons are made by **cosmic rays** about **10 km** up

They live **2.2 µs**: too short to reach the ground at c

But **many** reach the ground (Part 3)



Only Near the Speed of Light

Below 0.1c

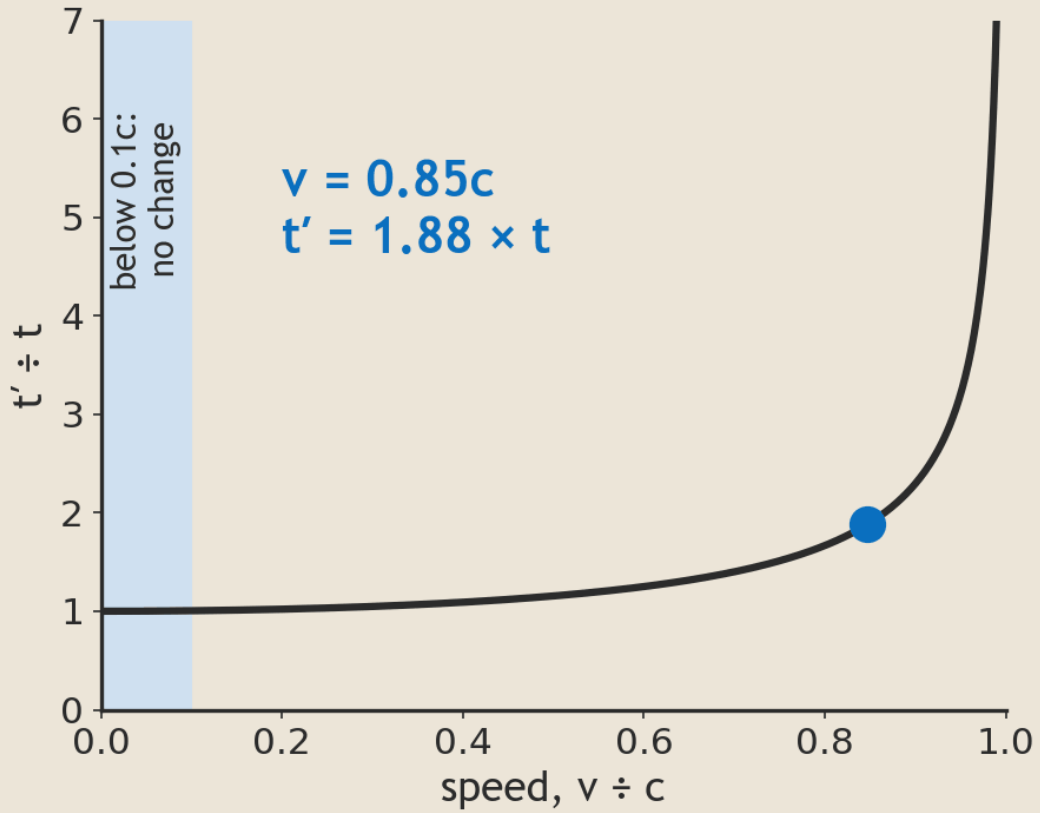
t′ is within **0.5 %** of t:
negligible

Trains, planes

are **not relativistic**

Near c

t′ grows **very fast**



Past Paper: 2025 Paper 2 Q4(c)

(c) A student makes the following statement regarding special relativity:

'Train drivers who travel at speeds of 80 $m\,s^{-1}$ over large distances will have to reset their watches frequently, as they spend so many hours travelling at high speeds.'

State whether the student is correct.

Justify your answer. **2**

[Turn over

Scan: page 15.

Plan: state first, then **justify**

Marking: 2025 Paper 2 Q4(c)

Question			Expected response		Max mark	Additional guidance
	(c)		The student is incorrect. The train does not travel at relativistic speeds.	**(1)** **(1)**	**2**	**JUSTIFY** Accept: Time dilation is negligible/ insignificant. **OR** No time dilation takes place. Accept: The train is not travelling at a speed close to the speed of light. The effects of special relativity only become significant when the speed $\geq 0.1c$. The train will not reach speeds of this magnitude.

Past Paper: 2016 Q4(b)(iii)

(b) On the plane, the students discuss the possibility of travelling at relativistic speeds.

(iii) One of the students states that the clocks on board the plane will run slower when the plane is travelling at relativistic speeds.

Explain whether or not this statement is correct. **1**

[Turn over

Scan: page 15.

Plan: **whose** frame? Say it.

Marking: 2016 Q4(b)(iii)

Question			Answer	Max Mark	Additional Guidance
		(iii)	Correct – from the perspective of the stationary observer there will be time dilation Incorrect – from the perspective of the students they are in the same frame of reference as the clock Not possible to say/could be both correct and incorrect – frame of reference has not been defined	1	The response must involve a statement referring to, or implying, a frame of reference

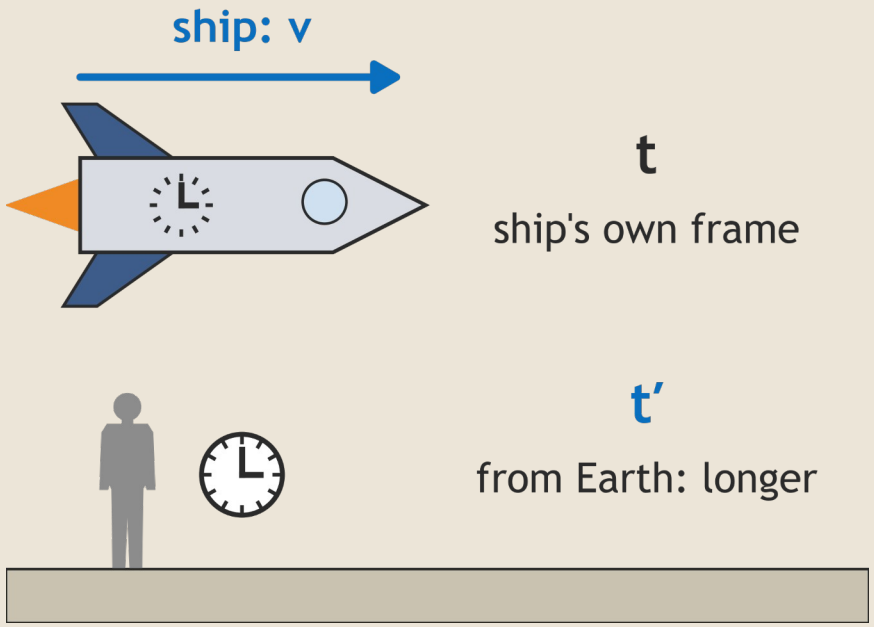
What Candidates Get Wrong (from course reports)

Dilation, not 'dilatation' (2025 report Q4(a))

Trains are **not relativistic** (2025 MI Q4(c), 71 %)

Clock answers refer to a **frame** (2016 MI Q4(b)(iii))

t and t' swapped: t is the **own frame** (teacher evidence)



Look back — by the end of this part…



1. I can **describe** how a **light clock** shows **time dilation**
2. I can **use** $t' = t / \sqrt{(1 - (v/c)^2)}$ to find **t'** or **t**
3. I can **state** that **moving clocks run slow** for a stationary observer
4. I can **explain** why **everyday speeds** show **no** time dilation

Course specification — 1 of 1

- Knowledge that measurements of space, time and distance for a moving observer are changed relative to those for a stationary observer, giving rise to time dilation and length contraction.
- Use of appropriate relationships to solve problems involving time dilation, length contraction, and speed.

Scan: the course specification, p.8–9.

Credits (1 of 2)

Title: caesium clock: Photo: Binarysequence, CC BY-SA 3.0, via Wikimedia Commons

Past Paper: 2025 Paper 2 Q4(a)(b): © Qualifications Scotland (SQA). 2025 Higher Physics Paper 2, Q4

Marking: 2025 Paper 2 Q4(a)(b): © Qualifications Scotland (SQA). 2025 Higher Physics Paper 2, Q4

Past Paper: 2024 Q7: © Qualifications Scotland (SQA). 2024 Higher Physics Paper 1, Q7

Past Paper: 2023 Q7: © Qualifications Scotland (SQA). 2023 Higher Physics Paper 1, Q7

Past Paper: 2021 Q10: © Qualifications Scotland (SQA). 2021 Higher Physics Paper 1, Q10

Evidence: Atomic Clocks 1: Photo: Binarysequence, CC BY-SA 3.0, via Wikimedia Commons

Evidence: Atomic Clocks 2: NASA/JPL-Caltech

Evidence: Muons: Photo: Cloudylabs, CC BY-SA 3.0, via Wikimedia Commons

Credits (2 of 2)

Past Paper: 2025 Paper 2 Q4(c): © Qualifications Scotland (SQA). 2025 Higher Physics Paper 2, Q4

Marking: 2025 Paper 2 Q4(c): © Qualifications Scotland (SQA). 2025 Higher Physics Paper 2, Q4

Past Paper: 2016 Q4(b)(iii): © Qualifications Scotland (SQA). 2016 Higher Physics Section 2, Q4

Marking: 2016 Q4(b)(iii): © Qualifications Scotland (SQA). 2016 Higher Physics Section 2, Q4

Course specification — 1 of 1: © Qualifications Scotland (SQA). Higher Physics Course Specification v4.1 (September 2026), p.8–9

R Stewart - November 2026